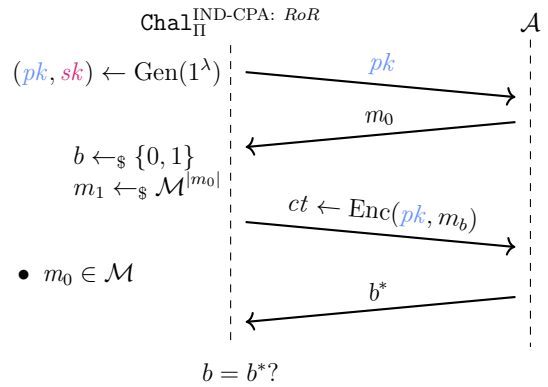
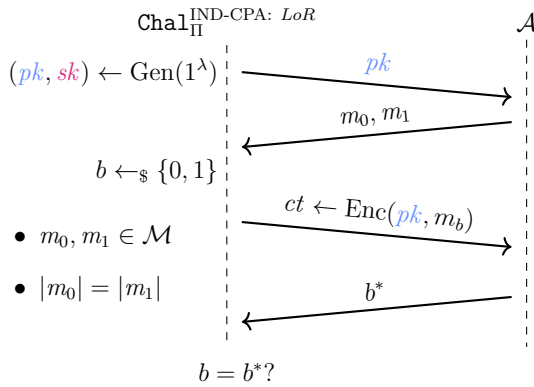


Exercise 1 (Negligible Functions). Prove or disprove that the following functions are negligible. You can use that $f(n) = \frac{1}{2^n}$ is negligible.

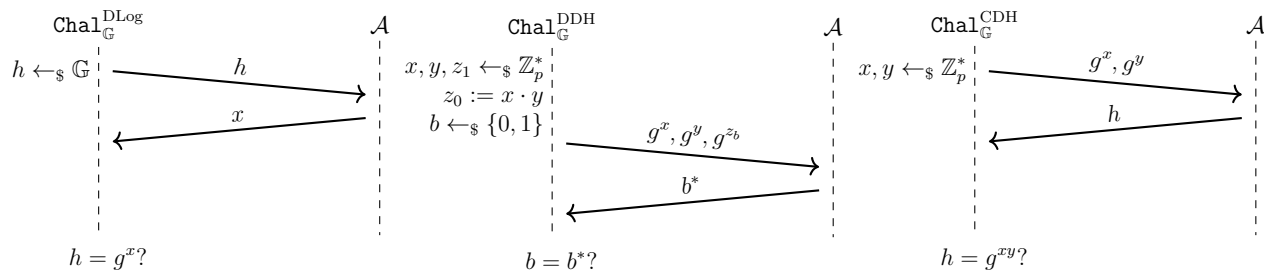
1. $f_1(n) = \frac{1}{n^{20}}$
2. $f_2(n) = \frac{2^n}{(2^n)^2}$
3. $f_3(n) = \left| \frac{(-1)^n}{2^n} \right|$
4. $f_4(n) = \frac{(-1)^n}{2^n}$
5. $f_5(n) = \frac{n^2 + n^7}{2^n}$

Exercise 2 (IND-CPA). Prove that Left-or-Right (LoR) IND-CPA (security game on the left) is equivalent to Real-or-Random (RoR) IND-CPA (security game on the right).



Exercise 3 (Perfect Secrecy). Prove that perfect secrecy is impossible for PKE schemes.

Exercise 4 (DLog, DDH, and CDH). Let $\mathbb{G}$ be a group with generator g and modulus p . Recall the security games for DLog (left), DDH (middle), and CDH (right):



1. Show that if DDH is hard in $\mathbb{G}$, then so is CDH.
2. Show that if DDH is hard in $\mathbb{G}$, then so is DLog.
3. Show that if CDH is hard in $\mathbb{G}$, then so is DLog.